

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-a41db3e368cc63da0.jsonl`  
- SHA-256: `d031ff3a83b8960c27e6d485b9ac10c84b73e576a833982f5fc13ec1c90a013f`  
- Source modified: `2026-07-02T02:53:21+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:47:11.979Z)

CONTEXT (read carefully):

- You are one auditor in a multi-agent tech-debt/critical-fix audit of the khelsutra-guru multi-repo workspace at C:/Users/avidu/Projects/khelsutra-guru. Today is 2026-07-02. All git repos were just pulled to current origin master/main.
- Repos: badminton-highlight-indexer (the ENGINE, Python, ~1500 tests, org Khelsutra), khelsutra (SPA web/ + marketing site/, TS/React, org avidullu), sports-data-collector, rally-annotator, sports-obsreport (small Python repos). Plus two NON-git dirs: "Annotation Setup", khelsutra-screencast.
- HARD RULES: READ-ONLY. Do NOT modify/create/delete any file, do NOT create branches/worktrees, do NOT pip install, do NOT run the full test suite (CI is trusted on master). You MAY run read-only git and gh commands (gh is authenticated), ruff/mypy IF already installed, and fast read-only scripts.
- EVERY claim must cite file:line (or a gh URL). A claim without file:line evidence is a hypothesis - mark it as such or drop it. The codebase drifts between sessions: re-verify anything you believe from docs against actual code.
- OWNER-GATED areas (flag as owner_gated=true, still report): alpha/serving go-live + Cloudflare dashboard work, Studio P10/P11 (corpus promotion + train/reeval), product/strategy/licensing decisions, real GPU/cloud spend, counsel ToS/DPA, repo rename off "badminton".
- Known-open items from project memory (VERIFY current state, don't trust blindly): engine PR #390 (D1 split main.py, open since 06-26) · audit doc docs/CODE_CLEANUP_AUDIT_2026-06-24.md remaining items D1, D13 (ByteTrack->trackers), Phase-3 (B3/B4/B6/B7/B8/D5/D6/D8-D10/O3/O4) · docs/DECISION_SNAPSHOT_REGRESSION.md next=P1 · docs/GOLDEN_REGRESSION_FIXTURES.md backlog M4/P0-rename/O1/O2 · engine issues #340 (Gemini re-bill), #332 (NSFW preflight), #349 (CPU-bound decode), #384 (CI single-source) · khelsutra issues #97-#114 UX/alpha backlog · DO droplet RETIRED 2026-06-25 (docs may still reference it; api.khelsutra.guru CNAME/CF-Access cleanup was pending).
- Categorize each finding: critical-fix (latent bug/correctness/data-loss/cost-leak), security, tech-debt (structure/duplication/dead code), test-gap, ci-infra, doc-rot, hygiene.
- Severity: P0 = latent bug or cost/data/security risk reachable in normal use; P1 = high-value debt blocking velocity or a stale-open thread; P2 = worthwhile cleanup; P3 = nice-to-have (report max 3 of these).
- Be selective: max ~12 findings, each genuinely worth acting on. Note tracked_in when an issue/PR/doc already tracks it (tracked is NOT done - still report). Your final message is data for an orchestrator, not prose for a human.

YOUR DIMENSION: structural tech debt in the ENGINE. Repo: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer.

Measure and report: backend/main.py size and responsibility count (D1 - check open PR #390's approach via gh pr view 390 --json files,title,body and whether master drifted under it); other god modules (wc -l over backend/**/*.py, flag >800 lines); duplicated logic across segmenters/runners (backend/pipeline/); dead code (vendored ByteTrack vs the 'trackers' package

- audit item D13 - find the vendored dir and its usage sites); import cycles; scratch/ or artifacts/ files that leaked into git (git ls-files | grep -E 'scratch|artifacts'); stale local

branches/worktrees (git branch -a | wc -l, git worktree list). Prioritize by what actually slows down changes.

2. user (2026-07-02T02:47:20.197Z)

e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)

master...origin/master

3. user (2026-07-02T02:47:22.875Z)

```
28907 total
2198 backend/main.py
1259 backend/infrastructure/database.py
1035 backend/eval/golden_fixtures.py
 870 backend/pipeline/detectors/wasb_infer.py
 849 backend/eval/rally_seg_eval.py
 759 backend/eval/experiment.py
 748 backend/worker/__main__.py
 694 backend/pipeline/segmenters/gemini.py
 599 backend/eval/ablation.py
 596 backend/pipeline/detectors/tracknet_runner.py
 558 backend/pipeline/segmenters/yolo_hybrid.py
 514 backend/pipeline/stitcher/compiler.py
 496 backend/config/models.py
 485 backend/pipeline/segmenters/trajectory_hybrid.py
 426 backend/pipeline/detectors/person_detector.py
 397 backend/eval/golden_real_fixtures.py
 373 backend/eval/gemini_refine.py
 340 backend/eval/tolerance_metrics.py
 328 backend/pipeline/detectors/native_wasb_runner.py
 324 backend/eval/per_user_regression.py
 324 backend/config/startup.py
 323 backend/eval/export_wasb_segments.py
 323 backend/eval/eval_partial_labels.py
 291 backend/config/loader.py
 263 backend/api/job_worker.py
 262 backend/utils/run_telemetry.py
 262 backend/pipeline/detectors/fusion_features.py
 261 backend/reporting/harvest.py
 261 backend/providers/gemini.py
 261 backend/eval/fusion_golden.py
 253 backend/eval/segmentation_metrics.py
 250 backend/eval/regression.py
 245 backend/eval/windowing.py
 243 backend/pipeline/segmenters/motion.py
 241 backend/pipeline/segmenters/fusion_hybrid.py
 236 backend/eval/ablation_pdf.py
 228 backend/pipeline/detectors/base.py
 228 backend/compute/cloud_run.py
 227 backend/tools/rally_slicer.py
```

4. user (2026-07-02T02:47:33.343Z)

```
{"createdAt":"2026-06-26T14:28:06Z","fileCount":3,"files":["backend/main.py","backend/orchestration.py","mypy.ini"],"state":"OPEN","title":"refactor(D1/P10): extract orchestration.py from main.py (god-module split, step 1)","updatedAt":"2026-06-26T14:37:18Z"}
```

5. user (2026-07-02T02:47:33.823Z)

```
* master
remotes/origin/HEAD -> origin/master
remotes/origin/chore/teardown-accept-rally-vm-name
remotes/origin/docs/nightly-person-cue-lane
remotes/origin/docs/preserve-rally-detection-review
remotes/origin/fix/rally-detection-edge-cases
remotes/origin/master
remotes/origin/refactor/p10-split-main
---
```

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer e410136 [master]

6. user (2026-07-02T02:47:35.092Z)

```
deploy/digitalocean/mount_scratch.sh
scratch/analyze_motion.py
scratch/clean_state.py
scratch/diagnose_rally.py
scratch/gemini_video_debugger.py
scratch/list_models.py
scratch/test_ai_segmenter.py
scratch/test_thresholds.py
---count---
```

8

7. user (2026-07-02T02:47:36.007Z)

8. user (2026-07-02T02:47:52.727Z)

```
{"baseRefName":"master","headRefId":"e59638ecb376b7570c4245a0bab92e9d4ea65357","mergeStateStatus":"DIRTY","mergeable":"CONFLICTING"}
---
```

b6214e58570b92de9558f8f8bd12f20d2f56f49a

9. user (2026-07-02T02:47:53.111Z)

```
9
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
52d97b5 P4: POST /api/assets ? store a video into the user's chosen medium (#409)
7ea6ea9 P3: /api/capabilities reports usable storage backends (#408)
f28af73 P2: engine free-space guard (HTTP 507 + WSL frame-stage BLOCK) (#407)
759a67f feat(api): frame-preserving annotation proxy for /api/source (D4) (#406)
5a5c4fa feat(api): P8a ? GET /api/source + POST /api/annotations (annotate-in-UI backend) (#405)
959c630 fix(#397): write per-video report to the resolved local path, not the raw URI (#400)
b6214e5 chore: ruff format the whole repo (259 files) (#387)
```

10. user (2026-07-02T02:47:54.022Z)

```
./backend/config/models.py:167: yolo_tracker: str = "bytetrack.yaml"
./backend/pipeline/detectors/person_detector.py:1: ""Person-detection rally cue ? torchvision
detector + ByteTrack + motion windowing.
./backend/pipeline/detectors/person_detector.py:11:exception is the ByteTrack swap (D13/P12):
`make_tracker` now builds a
./backend/pipeline/detectors/person_detector.py:12:`trackers.ByteTracker` with its
constructor params pinned to `sv.ByteTrack`'s
./backend/pipeline/detectors/person_detector.py:15: - `make_tracker` ? a fresh
ByteTrack tracker per chunk (standalone `trackers` pkg)
./backend/pipeline/detectors/person_detector.py:26:Licensing: torchvision detectors are BSD +
COCO weights; ByteTrack is MIT (via the
./backend/pipeline/detectors/person_detector.py:27:standalone `trackers` package, Apache-2.0,
since D13/P12; was `supervision.ByteTrack`)
./backend/pipeline/detectors/person_detector.py:58: Owns the torchvision model + per-chunk
```

```

ByteTrack tracker. Constructed with the run
./backend/pipeline/detectors/person_detector.py:106:
for one chunk (track IDs are chunk-local).
./backend/pipeline/detectors/person_detector.py:108:
``supervision.ByteTrack`` (deprecated, removed in 0.30.0)
./backend/pipeline/detectors/person_detector.py:109:
package (``ByteTrackTracker``). Imported lazily so
./backend/pipeline/detectors/person_detector.py:112:
``sv.ByteTrack``'s behaviour so the migration
./backend/pipeline/detectors/person_detector.py:124:
ByteTrackTracker
./backend/pipeline/detectors/person_detector.py:127:
./backend/pipeline/detectors/person_detector.py:142:
./backend/pipeline/detectors/person_detector.py:165:
be populated (it filters on it internally).
./backend/pipeline/detectors/person_detector.py:325:
local ByteTrack tracker is used. ? cv2 frame-seek
./backend/pipeline/detectors/person_detector.py:381:
``progress_cb(frames_done, total_frames)`` (called every ~256
./backend/pipeline/segmenters/yolo_hybrid.py:1:
"""Two-stage rally segmenter: torchvision person
detection + ByteTrack tracking
./backend/pipeline/segmenters/yolo_hybrid.py:13:
supervision's MIT-licensed ByteTrack.
"""Create a fresh ByteTrack tracker

D13/P12: migrated from

to the standalone ``trackers``

The constructor params are pinned to

from trackers import

return ByteTrackTracker(
ByteTrack does the association.
# ByteTrack requires confidence to

"frame_height"}``. A fresh window-

window-local ByteTrack.

```

11. user (2026-07-02T02:48:08.722Z)

```

18
---routes
18
---imports-top
1:import argparse
2:import csv
3:import json
4:import logging
5:import math
6:import os
7:import shutil
8:import socket
9:import sys
10:import tempfile
11:import threading
12:import uuid
13:import webbrowser
14:from importlib.metadata import PackageNotFoundError
15:from importlib.metadata import version as _pkg_version
16:from urllib.parse import quote
18:import uvicorn
19:from dotenv import load_dotenv
24:# stdlib-only import ? keeps this pre-pydantic line light.
25:from backend.utils.app_home import env_file_to_load, resolve_output_dir
36:from typing import Any, Dict, List, Optional, Tuple
38:from fastapi import FastAPI, HTTPException, Request
39:from fastapi.middleware.cors import CORSMiddleware
40:from fastapi.responses import FileResponse, HTMLResponse, JSONResponse
41:from fastapi.staticfiles import StaticFiles
42:from pydantic import BaseModel
43:from starlette.middleware.trustedhost import TrustedHostMiddleware
45:from backend import (
49:from backend.api import (
52:from backend.config import (

```

12. user (2026-07-02T02:48:10.009Z)

```

"""Two-stage rally segmenter: torchvision person detection + ByteTrack tracking
(Stage 1, local) feeding an AI provider for semantic boundaries (Stage 2).

```

Stage-1 detection now lives in `backend/pipeline/detectors/person\_detector.py`
(`PersonDetector`, a reusable rally-cue producer ? fusion P1) so the SAME person cue
can be consumed by the future fusion segmenter; this segmenter just orchestrates
chunking, the AI-provider handoff, and DB population around it. Behaviour is unchanged.

Stage 1 originally used the AGPL-licensed ultralytics YOLO. To keep the hosted
service free of copyleft obligations (see docs/MONETIZATION_AUDIT.md and ADR-005),
detection uses torchvision's permissively-licensed (BSD) detectors and the
association/tracking that YOLO.track() used to bundle in is provided by
supervision's MIT-licensed ByteTrack.

The registry key stays "yolo_hybrid" for back-compat with existing configs and the
DB `detector` tags, even though no YOLO is involved anymore.
"""

```
import concurrent.futures
import logging
import os
import time
from typing import Any, Dict, List, Tuple

from backend.pipeline.detectors.person_detector import PersonDetector
from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
from backend.providers import provider_registry
from backend.sports import sport_registry
from backend.utils.paths import output_base
from backend.utils.video import extract_video_chunk, get_video_duration
=====
1:"""Two-stage rally segmenter: torchvision person detection + ByteTrack tracking
13:supervision's MIT-licensed ByteTrack.
57:         "Two-stage pipeline: torchvision person detection + ByteTrack for fast "
102:         f"Initializing Hybrid Segmenter (torchvision + ByteTrack) for video:
{video_path}"
```

13. user (2026-07-02T02:48:11.230Z)

```
./artifacts/cloudrun_e2e_validation/RUN_TRANSCRIPT.md:118:python _scratch/make_reel.py
halfB_intervals.csv source_clip.mp4 . cloudrun_e2e_reel.mp4
./artifacts/cloudrun_e2e_validation/RUN_TRANSCRIPT.md:140:- `_scratch/make_reel.py` ? the local
stitch script.
./backend/eval/ablation.py:3:A *systematic, repeatable* version of the one-off ablation
(`_scratch/gate_ab_ablation.py`):
./backend/eval/ablation.py:6:95% CI + paired sign test + the C4 over-segmentation ratio. See
`_scratch/ABLATION_FRAMEWORK_PLAN.md`
./backend/eval/ablation.py:443: ? exactly the historical measurement
(`_scratch/gate_ab_ablation.py`)."""
./backend/eval/ablation_pdf.py:30: structural-protected). Reuses the muted booktabs aesthetic
of `_scratch/quality_report.py`.
./backend/eval/windowing.py:11:(`_scratch/GATE_AB_NUMBERS_AND_IMPLICATIONS.md` vs
`_output/gateA_served_verify/`).
./backend/utils/paths.py:8: """The configured base dir for a segmenter's LOCAL
scratch/staging artifacts
./docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md:95:> polls. See
`_scratch/PR1_HANDOFF.md` for the review bundle.
./docs/archives/past_projects/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md:48:1. `rclone copyto
$VIDEO_URI /scratch/in.mp4` ? **sync-first, not a FUSE random-access decode.**
./docs/archives/quality-iterations/2026-06-18-lovo-resweep-n13/lovo_resweep.py:9:Reproduce: add
any new clip via scratch/ingest_golden.py, then run this. The sys.path insert
./docs/archives/quality-iterations/2026-06-18-lovo-resweep-n13/README.md:59: via
`_scratch/ingest_golden.py` first). It self-reports the cap sweep + LOVO + R4 + significance.
./docs/archives/quality-iterations/2026-06-first-ab-experiment/README.md:28:- IoU 0.5. Repro:
`_scratch/ab_n6_lovo.py` (machine-side; reads the gitignored manifest + the external
./docs/CODE_MAP.md:34:| A throwaway repro / debug script | `_scratch/` | **Gitignored** +
excluded from ruff/pytest. Never imported by app code. |
./docs/DATA_IN_GCS.md:148:`trajectories/` + `labels/`). Sources (resolved 2026-06-22 ? see
```

```
`scratch/copy_videos_to_gcs.ps1`):
./docs/DATA_IN_GCS.md:160:- **Interim (today, owner box ? streams Drive/F: ? local ? GCS):**
`pwsh scratch/copy_videos_to_gcs.ps1
./docs/DATA_IN_GCS.md:198:> `scratch/`; promote it to `scripts/` if corpus-sync recurs).
./docs/DATA_IN_GCS.md:205:> script (`scratch/copy_videos_to_gcs.ps1`) stages such paths through
a bracket-free temp file; the
./docs/DATA_IN_GCS.md:230:5. **Reproducibility:** the migration + copy helpers live in
gitignored `scratch/`
./docs/data_pipeline/GOLDEN_VIDEOS.md:23: (`gts` + candidates), append a row to
`output/human_lovo_manifest.json`. (`scratch/at_par.py` scores a
```

14. user (2026-07-02T02:48:28.719Z)

```
6b7d5dd Refactor Gemini prompt to file top, add defensive timestamp parsing, add compiler
logging, and create scratch debug/cleanup tools
0f76429 Refactor rally detection to pluggable base and registry, integrate Gemini 2.5 Flash
segmenter, add local fast ffmpeg trimming, and update developer docs
72f972e Checkpoint: Add debug trim feature, runner script, and motion analysis scratch script
---gitignore
52:# Local generated artifacts (review reels, eval CSVs, e2e/proof outputs) -- regenerable,
never committed
53:artifacts/
69:# Throwaway repro scripts + PR-body / handoff scratch notes (already excluded from
71:scratch/
---artifacts-tracked
0
.gitignore:53:artifacts/          artifacts/cloudrun_e2e_validation/RUN_TRANSCRIPT.md
---ruff
```

15. user (2026-07-02T02:48:29.299Z)

```
50
6
---sections
177:# --- P2: the friendly `indexer --app` launcher (cross-platform, no native component ? 01)
---
312:# --- API Models ---
378:# --- API Endpoints ---
590:# --- Annotate-in-UI (P8): serve a video's source + persist finished rally labels
-----
1479:# --- Chunked upload (B2, PR-2) -----
1489:# --- Highlight download (B3, PR-2) -----
1709:# --- CLI Debug Utility & Orchestration ---
```

16. user (2026-07-02T02:48:30.038Z)

```
40:| **D1** | **Phase 1 = pure refactorings only.** No logic or behavior change. All tests must
pass; coverage must not decrease. | Owner, 2026-06-24 | B2 (Success/Failure contract for Gemini)
is deferred to Phase 2 since it changes return types callers depend on. D3 (typed-config
migration), D2 (app-factory), and B5 (segmenter validator) are pure-refactoring-possible and are
the first targets. |
46:| **D7** | **Absorb relevant BACKLOG items.** Deferred hardening items from `docs/BACKLOG.md`
that overlap with this audit are absorbed into this project rather than left in BACKLOG. Items
that are feature-gaps (not code-cleanup) stay in BACKLOG. | PR #352 review, 2026-06-24 | D2 and
01 were already independently found; D13 and D14 are new absorptions. The train/eval splits
guardrail stays in BACKLOG (it needs a collector-side manifest change). |
69:| **D1** | HIGH | `backend/main.py` | **Massive god-module** (~1450 lines). CLI, API, worker
orchestration, upload, debug-clip, trim, reingest all in one file. | 2 |
81:| **D13** | MEDIUM | `requirements.txt`, `pyproject.toml` | **Absorbed from BACKLOG.**
`supervision` ByteTrack is deprecated as of 0.28.0 and **removed in 0.30.0**. The `<0.30.0` pin
in `requirements.txt` and `pyproject.toml` is a stopgap ? migration to the standalone `trackers`
package is needed before the pin can be lifted. | 2 |
115:| **D13** | "`supervision` ByteTrack deprecation" | ? ? pinned `<0.30.0` as stopgap;
migration to `trackers` package needed | New finding in this doc (see S3b) |
```

168:- ****Does NOT split `main.py` (D1).**** The god-module split is deferred ? D2 (app-factory) is a necessary precursor.

198:| ****P10**** | ****D1: Split `main.py` god-module**** (~1450 lines ? smaller modules) | P7 | No | ? | ? |

200:| ****P12**** | ****D13: ByteTrack deprecation**** ? migrated `sv.ByteTrack` ? `trackers.ByteTrackTracker` (git dep; PyPI wheel broken) | P7 | No | ? |

[#386](https://github.com/Khelsutra/badminton-highlight-indexer/pull/386) |

211:1. ****Q: Should D1 (split `main.py`) be Phase 1 or Phase 2?**** ? Leaning Phase 2: D2 (app-factory) is a natural precursor, and D1 is lower risk after the globals are gone. Owner to confirm.

221:- [x] P5: `create\_app()` injects shared state via `app.state` (handlers read `request.app.state`); `backend.api.job\_worker` removed from mypy quarantine. (The Optional module globals `db\_instance`/`global\_config` are ****retained as a transitional back-compat shim**** and `backend.main` stays mypy-quarantined ? full removal is follow-up, see D1/P10.)

246:| ~4~ | ~**D13 / P12**~ | ~ByteTrack deprecation migration~ ? ****? DONE.**** `sv.ByteTrack` ? `trackers.ByteTrackTracker` (git dep; PyPI wheel broken ? filed Roboflow issue). |

247:| 5 | ****D1 / P10**** | Split `main.py` god-module | ~1450 lines. D2 (app-factory) was the necessary precursor ? now done. Extract: CLI handlers ? `backend/cli/`, upload routes ? already in `backend/api/uploads.py`, debug helpers ? `scripts/`. |

261:- ****Audit source:**** 33 findings from a deep whole-repo read on 2026-06-24, plus 2 absorbed from `docs/BACKLOG.md` (D13, D14) for a total of 35. Two additional BACKLOG items (D2, O1) were already independently found and are cross-referenced in §3e.

273:- `2026-06-24` ? Absorbed 2 findings from `docs/BACKLOG.md` (D13, D14). Cross-referenced D2 and O1 in §3e.

282:- `2026-06-26` ? ****P12 CI fix (#386).**** The D13/P12 PR failed the `full-suite` lane with `ModuleNotFoundError: No module named 'trackers'`. Two root causes: (1) the two new tests hard-imported the standalone `trackers` package, but it's a git-only dep (PyPI wheel is metadata-only ? roboflow/trackers#474) deliberately omitted from `pyproject.toml`, and the lane installs via `pip install -e .[dev]`; (2) the git ref `@v2.5.0` was wrong ? roboflow/trackers uses bare version tags (`2.5.0`), so pip failed with `pathspec 'v2.5.0' did not match`. Fixes: added a `pip install trackers @ git+?@2.5.0` step to both `ci.yml` and `nightly.yml` full-suite lanes (same out-of-band pattern as torch); fixed the tag in `requirements.txt`; and guarded the package-importing test with `pytest.importorskip("trackers")` (matching the repo's `obsreport` soft-dep convention). All 10 CI checks green.

17. user (2026-07-02T02:48:51.041Z)

390 refactor/p10-split-main 2026-06-26T14:28:06Z refactor(D1/P10): extract orchestration.py from main.py (god-module split, step 1)

18. user (2026-07-02T02:48:52.340Z)

```
cloudrun
digitalocean
gcp
---
deploy/cloudrun/Dockerfile
deploy/cloudrun/README.md
deploy/cloudrun/cloudbuild.yaml
deploy/digitalocean/README.md
deploy/digitalocean/bootstrap.sh
deploy/digitalocean/gpu_setup.sh
deploy/digitalocean/mount_scratch.sh
deploy/gcp/README.md
deploy/gcp/buckets.env
deploy/gcp/create_gpu_vm.sh
deploy/gcp/nightly_regression/Dockerfile.launcher
deploy/gcp/nightly_regression/README.md
deploy/gcp/nightly_regression/cloudbuild.launcher.yaml
deploy/gcp/nightly_regression/create_nightly_bucket.sh
deploy/gcp/nightly_regression/launch.sh
deploy/gcp/nightly_regression/register_scheduler.sh
deploy/gcp/nightly_regression/run_on_vm.sh
deploy/gcp/nightly_regression/setup_email.sh
```

```
deploy/gcp/ops-agent-startup.sh
deploy/gcp/provision.sh
---ruff-exclude
---pytest
```

19. user (2026-07-02T02:49:02.682Z)

```
CYCLE: backend.config.models <-> backend.pipeline.segmenters.base
CYCLE: backend.eval.annotations <-> backend.eval.gt_loader
CYCLE: backend.api.job_worker <-> backend.api.uploads <-> backend.main
CYCLE: backend.eval.ablation <-> backend.eval.ablation_pdf
CYCLE: backend.eval.golden_fixtures <-> backend.eval.golden_real_fixtures
```

20. user (2026-07-02T02:49:15.203Z)

```
backend/api/job_worker.py:3:Extracted verbatim from backend.main to shrink the orchestrator;
behavior is unchanged.
backend/api/job_worker.py:11:`backend.main` module object at call-time (`import backend.main as
_main`), never as a
backend/api/job_worker.py:12:bare local. `backend.main` re-exports these names, so
`monkeypatch.setattr(backend.main,
backend/api/job_worker.py:14:`_arm_wake_timer`, so no real daemon timer leaks in tests).
Importing `backend.main` lazily
backend/api/job_worker.py:75:     import backend.main as _main
backend/api/job_worker.py:98:     import backend.main as _main
backend/api/job_worker.py:214:    import backend.main as _main
backend/api/job_worker.py:234:    import backend.main as _main
backend/api/job_worker.py:251:    import backend.main as _main
backend/api/uploads.py:1:"""Chunked upload (B2, PR-2) ? extracted from backend.main into its own
router.
backend/api/uploads.py:10:The startup config lives on ``backend.main.global_config`` (set ONLY
in ``main()``).
backend/api/uploads.py:11:``_upload_cfg`` resolves it through the ``backend.main`` module object
so it always
backend/api/uploads.py:13:``backend.main.global_config`` flow through unchanged).
backend/api/uploads.py:47:     import backend.main as _main
===
49:from backend.api import (
96:    ``backend.main`` and ``backend.api.job_worker`` into mypy quarantine.
286:# now lives in backend.api.job_worker. Re-exported here so the names stay addressable on
292:from backend.api.job_worker import ( # noqa: F401 (re-exports: names stay addressable on
backend.main)
1481:# backend.api.uploads as a self-contained APIRouter. It resolves the live startup config
via
1484:from backend.api import uploads as uploads_api
===
231:     # FULLY-LOCAL mode (default OFF): when true, the trajectory-hybrid segmenters
249:         from backend.pipeline.segmenters.base import segmenter_registry
260:         # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-positive
===
15:         from backend.config.models import AppConfig
```

21. user (2026-07-02T02:49:16.721Z)

```
docs/CLOUD_GPU_DRYRUN_GUIDE.md:80:     bash deploy/digitalocean/bootstrap.sh --gpu
# venv + CUDA torch
docs/CLOUD_GPU_DRYRUN_GUIDE.md:82:     bash deploy/digitalocean/gpu_setup.sh
# the wasb conda env + WASB-SBDT
docs/HOSTING_PLAN.md:32:     ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only
`cloudflared` service on the box)
docs/HOSTING_PLAN.md:54:                                     ??? api.khelsutra.guru ???
Cloud Run (FastAPI control plane, scale-to-zero)
docs/HOSTING_PLAN.md:78:| **All-DigitalOcean** | ~$11 ? ~$17 ($6 droplet + Spaces $5: 250 GiB +
**1 TiB egress included**) | One simple vendor; Spaces' included egress is the best video-
```



```

download deal; per-second billing | No credits; no scale-to-zero; still needs a tunnel-
equivalent to reach the box (run `frp`/WireGuard on the droplet) |
docs/NEXT_STEPS.md:33:15. Owner housekeeping: retire the shared **DO droplet** + **DO token**;
collector **md5** export (collector repo).
docs/README.md:25:- [SETUP_NEW_MACHINE.md](SETUP_NEW_MACHINE.md) ? **? The canonical "spin up a
box ? ready for training & inference" runbook.** Copy-pasteable, human-followable, **verified
end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see
[`deploy/gcp/README.md`](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite
(922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training,
with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/*` scripts are
provider-agnostic and reused on GCP.
docs/SETUP_NEW_MACHINE.md:17:> ? Assumes the DECOUPLED_COMPUTE PRs are merged: the CPU-mock
detector seam, `deploy/digitalocean/bootstrap.sh`,
docs/SETUP_NEW_MACHINE.md:66:doctl compute droplet create rally-box --image ubuntu-24-04-x64
--size s-2vcpu-4gb \
docs/SETUP_NEW_MACHINE.md:69:doctl compute droplet create rally-gpu --image gpu-h100x1-base
--size gpu-4000adax1-20gb \
docs/SETUP_NEW_MACHINE.md:71:doctl compute droplet list    # note the IP -> ssh -i ~/.ssh/rally
root@<IP>
docs/SETUP_NEW_MACHINE.md:100:./deploy/digitalocean/bootstrap.sh          # CPU box
docs/SETUP_NEW_MACHINE.md:102:./deploy/digitalocean/bootstrap.sh --gpu    # GPU box (installs
CUDA torch)
docs/SETUP_NEW_MACHINE.md:192: "endpoint_url": "https://sgp1.digitaloceanspaces.com", // DO
Spaces (VERIFIED); region = the Space's, e.g. sgp1
docs/SETUP_NEW_MACHINE.md:267:sudo bash deploy/digitalocean/mount_scratch.sh && export
TMPDIR=/scratch    # detect+mount the spare NVMe
===

```

Testing the rally pipeline on DigitalOcean

```

> **? DEPRECATED (ADR-013, 2026-06-20) ? DigitalOcean is DROPPED for compute.** Use
> **[`deploy/gcp/README.md`](../gcp/README.md)** ? GCP is the sole compute stack (DO GPU isn't
enabled
> on the account + isn't cleanly credit-funded; Paperspace is subscription-gated). Kept for
history.
> **NOTE:** the scripts here ? `bootstrap.sh`, `mount_scratch.sh`, `gpu_setup.sh` ? are
> **provider-agnostic** Linux-GPU setup and are **still reused by the GCP runbook**; only the
> DO-specific **provisioning** (doctl droplet create, Spaces) is retired. (Optional later cleanup:
move
> them to `deploy/common/`.)

```

A storage-free, secret-free way to get the pipeline running on a DigitalOcean (or Paperspace) machine ? the ****compute**** half of [docs/archives/past_projects/DECOUPLED_COMPUTE/](../docs/archives/past_projects/DECOUPLED_COMPUTE/README.md).

The object-store ****storage shim** is intentionally out of scope here****** (owner is finalizing it); this

runbook only proves ****compute**** works on the box.

22. user (2026-07-02T02:49:33.760Z)

1797

```

def _normalize_config(config: Any) -> Any:
    """Normalize a plain-dict config into an AppConfig so segmenters can use typed
    attribute access consistently. Production always passes AppConfig; tests may
    pass plain dicts ? this bridge keeps both paths working."""
    if isinstance(config, dict):
        from backend.config.models import AppConfig

        try:
            return AppConfig(**config)
        except Exception:
            return (

```

23. user (2026-07-02T02:49:35.316Z)

```

35: def _fmt_ts(seconds: float) -> str:
44:     def __init__(self, db, config):
51:     def name(self) -> str:
55:     def description(self) -> str:
61:     def _resolve_provider(self, video_path: str):
106:    def _collect_candidate_windows(
151:        def _extract_chunk(ci: int, cs: float) -> Tuple[int, float, str, bool]:
231:        def _merge_stats(a: Dict[str, Any], b: Dict[str, Any]) -> Dict[str, Any]:
262:    def _persist_candidates(
308:    def _run_ai_handoff(
334:        def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
404:    def _populate_db(
457:    def process_video(
====trajectory
40: def _fmt_ts(seconds: float) -> str:
56:     def _build_runner(
62:     def _build_native_runner(
73:     def _resolve_runner(
114:    def _probe_fps_width(video_path: str) -> tuple:
123:    def _shot_count_from(segment: Dict[str, Any], duration: float) -> int:
138:    def _enrich_candidate_stats(
158:    def _select_for_handoff(
171:    def process_video( # type: ignore[override]
390:        def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
====fusion
47:     def __init__(self, db: Any, config: Any):
57:     def name(self) -> str:
61:     def description(self) -> str:
68:     def _build_runner(
87:     def _build_native_runner(
111:    def _enrich_candidate_stats(
150:    def _axis_estimate(self, points: List[Any]) -> tuple:
161:    def _person_features(
199:    def _select_for_handoff(

```

24. user (2026-07-02T02:49:46.237Z)

```

backend/pipeline/segmenters/base.py:26: class BasePlaySegmenter(ABC):
backend/pipeline/segmenters/base.py:62: class SegmenterRegistry:
backend/pipeline/segmenters/fusion_hybrid.py:41: class
FusionSegmenter(TrajectoryHybridSegmenter):
backend/pipeline/segmenters/gemini.py:71: class GeminiPlaySegmenter(BasePlaySegmenter):
backend/pipeline/segmenters/motion.py:14: class MotionPlaySegmenter(BasePlaySegmenter):
backend/pipeline/segmenters/tracknet_hybrid.py:23: class
TrackNetHybridSegmenter(TrajectoryHybridSegmenter):
backend/pipeline/segmenters/trajectory_hybrid.py:47: class
TrajectoryHybridSegmenter(BasePlaySegmenter):
backend/pipeline/segmenters/wasb_hybrid.py:25: class
WasbHybridSegmenter(TrajectoryHybridSegmenter):
backend/pipeline/segmenters/yolo_hybrid.py:43: class HybridYoloSegmenter(BasePlaySegmenter):
===
def _fmt_ts(seconds: float) -> str:
    """Format seconds as HH:MM:SS for cross-verifying a timestamp against the video."""
    seconds = max(0, int(seconds))
    h, rem = divmod(seconds, 3600)
    m, s = divmod(rem, 60)
    return f"{h:02d}:{m:02d}:{s:02d}"

---
def _fmt_ts(seconds: float) -> str:
    seconds = max(0, int(seconds))
    h, rem = divmod(seconds, 3600)
    m, s = divmod(rem, 60)
    return f"{h:02d}:{m:02d}:{s:02d}"

```

```
class TrajectoryHybridSegmenter(BasePlaySegmenter):
```

25. user (2026-07-02T02:49:47.252Z)

```
def _build_runner(
    self, idx_cfg: IndexingConfig
) -> Tuple[DetectorRunner, Dict[str, Any]]:
    """Return (runner, detector-specific detection_params extras) for the default (WSL)
path."""
    raise NotImplementedError

def _build_native_runner(
    self, idx_cfg: IndexingConfig
) -> Tuple[DetectorRunner, Dict[str, Any]]:
    """Return (runner, extras) for the Linux-native (no-WSL) path. Only families with a
native runner override this; the base refuses with a clear message (M3.5: WASB only)."""
    raise NotImplementedError(
        f"detector_impl='native' is not supported for the "
        f"{self.display_label or self.family or 'trajectory'!r} family ? only WASB has a "
        f"Linux-native runner. Use detector_impl='auto' (WSL) or 'stub'."
    )

def _resolve_runner(
    self, idx_cfg: IndexingConfig
) -> Tuple[DetectorRunner, Dict[str, Any]]:
====fusion
def _build_runner(
    self, idx_cfg: IndexingConfig
) -> Tuple[DetectorRunner, Dict[str, Any]]:
    if isinstance(idx_cfg, dict):
        idx_cfg = IndexingConfig(**idx_cfg)
    substrate = idx_cfg.fusion.substrate
    if substrate == "tracknet":
        cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
        return TrackNetRunner(cfg), {
            "weights_path": cfg.weights_path,
            "queue_length": cfg.queue_length,
            "fusion_substrate": "tracknet",
        }
    wcfg = WasbConfig.from_indexing_cfg(idx_cfg)
    return WasbRunner(wcfg), {
        "weights_path": wcfg.weights_path,
        "fusion_substrate": "wasb",
    }

def _build_native_runner(
    self, idx_cfg: IndexingConfig
) -> Tuple[DetectorRunner, Dict[str, Any]]:
    # Native (no-WSL) path supports the WASB substrate only ? there is no native TrackNet
    # runner yet. Fusion's default substrate is WASB, so this covers the common case.
    substrate = idx_cfg.fusion.substrate
    if substrate != "wasb":
        raise NotImplementedError(
            "detector_impl='native' supports the WASB substrate only ? set "
            "indexing.fusion.substrate='wasb' (no native TrackNet runner yet)."
        )
    from backend.pipeline.detectors.native_wasb_runner import (
        NativeWasbConfig,
        NativeWasbRunner,
    )

    cfg = NativeWasbConfig.from_indexing_cfg(idx_cfg)
    return NativeWasbRunner(cfg), {
        "weights_path": cfg.weights_path,
        "detector_impl": "native",
```

```

        "fusion_substrate": "wasb",
        "device": cfg.device,
    }

```

26. user (2026-07-02T02:50:00.102Z)

```

diff: /c/Users/avidu/AppData/Local/Temp/y.txt: No such file or directory
===diff-lines===
diff: /c/Users/avidu/AppData/Local/Temp/y.txt: No such file or directory
0

```

27. user (2026-07-02T02:50:02.862Z)

```

backend/eval/golden_fixtures.py:950:          # The real-fixture subsystem (P1 + M2) lives in
golden_real_fixtures (split out); import it
backend/eval/golden_fixtures.py:952:          from backend.eval import golden_real_fixtures
backend/eval/golden_fixtures.py:1005:# is exposed here so every historical
``golden_fixtures.<name>`` import path stays valid.
backend/eval/golden_fixtures.py:1009:#: ``golden_fixtures`` ? ``golden_real_fixtures`` cycle
resolves regardless of import order ?
backend/eval/golden_fixtures.py:1010:#: ``golden_real_fixtures`` imports the shared primitives
FROM this module, so an EAGER bottom-of-
backend/eval/golden_fixtures.py:1011:#: file re-export would fail when ``golden_real_fixtures``
is imported FIRST (its own names would
backend/eval/golden_fixtures.py:1028:          from backend.eval import golden_real_fixtures
backend/eval/golden_real_fixtures.py:10:manifest appender) live in ``golden_fixtures`` and are
imported FROM it here ? there is a single
backend/eval/golden_real_fixtures.py:13:module so every historical ``golden_fixtures.<name>``
import path stays valid.
backend/eval/golden_real_fixtures.py:27:# --- shared primitives, imported FROM golden_fixtures
(single source of truth) -----
backend/eval/golden_real_fixtures.py:28:from backend.eval.golden_fixtures import (
===
backend/config/models.py:150:    default_segmenter: str = "yolo_hybrid"
tests/test_config.py:24:    assert cfg.indexing.default_segmenter == "yolo_hybrid"
tests/test_config.py:172:          "indexing": {"default_segmenter": "yolo_hybrid"},
tests/test_config.py:249:    assert cfg.indexing.default_segmenter == "yolo_hybrid"
tests/test_config_deployment.py:177:    assert cfg.indexing.default_segmenter == "yolo_hybrid"
# untouched default
tests/test_database.py:148:    "yolo_hybrid",
tests/test_database.py:161:    assert c["detector"] == "yolo_hybrid"
tests/test_database.py:171:    assert len(db.query_candidate_segments("vid1",
detector="yolo_hybrid")) == 1

```

28. user (2026-07-02T02:50:12.553Z)

```

141
===
1,26c1,12
<      def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
<          w_start = max(0, window["start"] - handoff_padding)
<          w_end = min(video_duration, window["end"] + handoff_padding)
<          w_dur = w_end - w_start
<
<          w_path = os.path.join(gemini_dir, f"handoff_{video_id}_{wi}.mp4")
<          # Re-encode short handoff clips to avoid keyframe-boundary drift
<          success = extract_video_chunk(
<              video_path, w_start, w_dur, w_path, reencode=True
<          )
<          if not success:
<              return []
<
<          prompt = sport_profile.get_prompt(chunk_duration=w_dur)
<          segments, metrics = provider.analyze_video(

```

```

<         w_path, prompt, chunk_duration=w_dur, label=f"Handoff {wi + 1}"
<     )
<     # B2: a provider failure returns [] with metrics["error"] set ? that is
<     # "we don't know", NOT "no rallies in this window". Surface it loudly so a

```

29. user (2026-07-02T02:50:13.429Z)

```

21:class SchemaTooNewError(RuntimeError):
25:class Database:
===methods
50

```

30. user (2026-07-02T02:50:24.298Z)

```

def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
    w_start = max(0, window["start"] - handoff_padding)
    w_end = min(video_duration, window["end"] + handoff_padding)
    w_dur = w_end - w_start
    w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
    if not extract_video_chunk(
        video_path, w_start, w_dur, w_path, reencode=True
    ):
        return []
    prompt = sport_profile.get_prompt(chunk_duration=w_dur)
    segments, metrics = provider.analyze_video(
        w_path, prompt, chunk_duration=w_dur, label=f"Handoff {wi + 1}"
    )
    # B2: a provider failure returns [] with metrics["error"] set ? that is
    # "we don't know", NOT "no rallies in this window". Surface it loudly so a
    # partial run isn't mistaken for a complete one (mirrors the gemini segmenter).
    if metrics.get("error"):
        logger.warning(
            "[Handoff %d] provider error ? window %d skipped: %s",
            wi + 1,
            wi,
            metrics["error"],
        )
    valid = []
    for seg in segments:
        try:
            seg["start_time"] = float(seg["start_time"]) + w_start
            seg["end_time"] = float(seg["end_time"]) + w_start
            seg["_candidate_id"] = candidate_ids[wi]
            valid.append(seg)
        except (ValueError, TypeError, KeyError) as e:
            logger.warning(
                "Dropping segment with unparseable timestamps: %s ? %s",
                seg,
                e,
            )
    try:
        os.remove(w_path)
    except OSError:
        pass
    return valid

with concurrent.futures.ThreadPoolExecutor(
    max_workers=ai_max_workers
) as executor:
    futures = [
        executor.submit(
            process_candidate_window, wi, all_candidate_windows[wi]
        )
        for wi in handoff_indices
    ]

```

31. user (2026-07-02T02:50:47.475Z)

```
Exit code 2
===ruffcfg
ruff.toml
50:     "pytest",
51:     "pytest-cov",
53:     "ruff",
65:# tests skip via pytest.importorskip. For local dev against a sibling checkout:
===
ruff.toml:8:# scratch/ = throwaway debugging scripts that may hit real APIs; it is also
ruff.toml:10:extend-exclude = ["scratch"]
```

32. user (2026-07-02T02:50:48.218Z)

```
import os
import json
from dotenv import load_dotenv

load_dotenv()
from backend.infrastructure.database import Database
from backend.pipeline.segmenters import segmenter_registry

def test_pipeline():
    video_path = r"C:\Users\avidu\Videos\TestLargeVideo.MP4"
    if not os.path.exists(video_path):
        print(f"[ERROR] Test video not found at: {video_path}")
        return

    # Load configuration
    if os.path.exists("config.json"):
        with open("config.json", "r") as f:
            config = json.load(f)
    else:
        config = {}

    # Debug: Trim to first 15 seconds to check 7s-12s rally detection
    if "indexing" not in config:
        config["indexing"] = {}
    config["indexing"]["debug_duration"] = 15.0

    # Initialize a temporary testing database
    os.makedirs("output", exist_ok=True)
    db_path = os.path.join("output", "test_indexer.db")
    if os.path.exists(db_path):
        ===

scratch/ab_eval_demo.py
scratch/ab_n6_lovo.py
scratch/analyze_19june.py
scratch/at_par.py
scratch/compare_19june_report.py
scratch/compare_local_vs_gemini.py
scratch/eval_multitrack_19june.py
scratch/eval_multitrack_gx128_golden.py
scratch/extract_person_19june.py
scratch/gate_ab_ablation.py
scratch/gemini_detect_19june.py
scratch/gemini_long_detect.py
scratch/make_boxhill_highlights.py
scratch/measure_gate_a.py
scratch/measure_served_gate_a_15clip.py
scratch/p3_smoke.py
scratch/r1_evidence.py
scratch/r5_evidence.py
scratch/run_gx010125.py
```

```
scratch/run_gx010125_full.py
scratch/run_gx010125_rescore.py
scratch/run_wasb_gx128.py
scratch/served_gateA_sweep.py
scratch/served_gateA_verify.py
scratch/stitch_min5.py
scratch/test_ai_segmenter.py
scratch/verify_highlights.py
```

33. user (2026-07-02T02:51:02.284Z)

```
===unmerged
  origin/chore/teardown-accept-rally-vm-name
  origin/docs/nightly-person-cue-lane
  origin/docs/preserve-rally-detection-review
  origin/fix/rally-detection-edge-cases
  origin/refactor/p10-split-main
===dates
chore/teardown-accept-rally-vm-name: 2026-06-23 17:08:59 +0000 chore(gcp): teardown.sh accepts
RALLY_VM_NAME (align with create_gpu_vm.sh
docs/nightly-person-cue-lane: 2026-06-26 13:45:13 +0000 docs(nightly): add person-cue Stage-1
stats lane to the engine-regression
docs/preserve-rally-detection-review: 2026-06-29 14:24:44 +0530 fix(rally): correct _spread
zero-collapse and _trim_inactive_tail sparse d
fix/rally-detection-edge-cases: 2026-06-29 14:38:12 +0530 fix(rally): gate finding-B density fix
behind default-OFF inactivity_tempo
refactor/p10-split-main: 2026-06-26 14:37:15 +0000 fix(D1/P10): re-export all 8 orchestration
symbols + quarantine orchestrat
```

34. user (2026-07-02T02:51:02.580Z)

```
1:from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
10:  "segmenter_registry",
===pkg
5:# reason), so the wheel target packages are listed EXPLICITLY below ? hatchling's
6-# auto-detection would otherwise miss the un-__init__'d `backend` root.
7-#
8-# torch/torchvision are deliberately NOT listed in [project.dependencies]:
9-# their CPU/CUDA wheels need an out-of-band index (--index-url / --extra-index-url),
10-# which PEP 621 cannot express. Install them separately ? see [project.optional-dependencies]
--
101:packages = ["backend", "training"]
```

35. user (2026-07-02T02:51:17.605Z)

```
== chore/teardown-accept-rally-vm-name
392 MERGED chore(gcp): teardown.sh accepts RALLY_VM_NAME (align with create_gpu_vm.sh)
351 MERGED chore(gcp): teardown.sh accepts RALLY_VM_NAME (align with create_gpu_vm.sh)
== docs/nightly-person-cue-lane
389 MERGED docs(nightly): add person-cue Stage-1 stats lane to the engine-regression plan
(design)
== docs/preserve-rally-detection-review
393 CLOSED fix(rally): _spread fix + flag-gated _trim_inactive_tail density fix (P0) + preserved
review docs
== fix/rally-detection-edge-cases
394 MERGED Fix/rally detection edge cases
```

36. user (2026-07-02T02:51:29.198Z)

```
{"files":["backend/config/models.py","backend/eval/windowing.py","backend/pipeline/detectors/rally_gate.py","backend/pipeline/detectors/tracknet_runner.py","backend/pipeline/segmenters/trajectory_hybrid.py","docs/NEXT_STEPS.md","docs/RALLY_DETECTION_CODE_REVIEW.md","docs/RALLY_DETECTION_FIXES_PLAN.md","tests/test_rally_gate.py","tests/test_tracknet_runner.py","tests/test_windowing.py"]}
```

===394

```
["backend/config/models.py", "backend/eval/windowing.py", "backend/pipeline/detectors/rally_gate.py", "backend/pipeline/detectors/tracknet_runner.py", "backend/pipeline/segmenters/trajectory_hybrid.py", "docs/NEXT_STEPS.md", "docs/RALLY_DETECTION_CODE_REVIEW.md", "docs/RALLY_DETECTION_FIXES_PLAN.md", "tests/test_rally_gate.py", "tests/test_tracknet_runner.py", "tests/test_windowing.py"]
```

37. user (2026-07-02T02:51:44.097Z)

```
14:# backend.api.job_worker lifted from quarantine (now takes explicit config/db params).
15:[mypy-backend.main]
16-ignore_errors = True
17-
18-# Worker dispatcher: composes the SAME validator/segmenter/report building blocks as
19:# backend.main, which are typed `config: dict` but receive the AppConfig migration shim
20:# at runtime (the .get() compat layer). Shares backend.main's config-shim quarantine
21-# until the typed-config migration removes the dict?AppConfig duality everywhere.
22-[mypy-backend.worker.__main__]
23-ignore_errors = True
===globals
98:     Also sets the module-level ``global_config`` / ``db_instance`` for transitional
101:     global global_config, db_instance
102:     global_config = config
103:     db_instance = db
279:db_instance: Optional[Database] = None
280:global_config: Optional[AppConfig] = None # now typed (Pydantic model)
1138:     module-level ``global_config``/``db_instance`` (backward-compatible with existing
1157:         config = global_config
===count
54
```

38. assistant (2026-07-02T02:52:29.719Z)

All measurements complete. Compiling the audit findings.

39. user (2026-07-02T02:53:21.339Z)

Structured output provided successfully